Contact Finder — Daily Summary & Analysis Report

Comprehensive Overview of Development Updates & Dataset Investigation

Date: Sept 9, 2026
Target File: investigation_results (2).csv

📌 Executive Summary

Today's development focused on resolving critical performance bottlenecks, fixing bulk processing count bugs, implementing abort controls, and auditing search engine resource costs. An in-depth analysis of **2,165 records** in investigation_results (2).csv was conducted, revealing an **87.5% contact hit rate** on processed samples, alongside actionable paths to increase confidence scores and optimize search runtime from 7.5 minutes down to ~1-2 minutes.

1. Summary of Today's Development Work

🐛 Bug Fixes & Engine Stability

- **Bulk Processing Match Counter Fix:** Resolved logic issue where bulk processing completed successfully but falsely reported Matches: 0.
- **Abort Controller Integration:** Added "Stop" capability to single and batch search flows using standard AbortController signals, allowing safe user cancellation without data loss.
- **Instant Cache Hits:** Fixed a bug where cached records were still subjected to a 25-second rate-limiting throttle delay. Cache hits now resolve instantly in milliseconds.

🎨 UI & Branding Enhancements

- **Favicon Overhaul:** Designed and implemented a vector-based, borderless transparent SVG favicon (icon.svg), replacing legacy bordered assets.

⚡ Performance & API Cost Optimization

- **Parallel Deep Search Engine:** Parallelized the search, fetch, and extraction pipeline inside CoordinatorAgent.ts for multi-threaded processing.
- **ScrapingBee vs. Puppeteer Audit:** Identified that ScrapingBee burned ~150–200 credits per person (exhausting 1,000 credits in ~5 searches) due to 15 queries/person and forced JS rendering (5x multiplier).
- **Zero-Cost Architecture:** Safely rolled back the engine to local Puppeteer (BrowserFetcher.ts) operating 3 parallel headless tabs, maintaining high speed with **$0 API overhead**.

2. Results Analysis: investigation_results (2).csv

TOTAL RECORDS	PROCESSED SAMPLE	PHONE HIT RATE	AVG CONFIDENCE

2,165	64	87.5%	26.41%

Confidence Score Tier Breakdown

Confidence Tier	Count	Percentage	Characteristics & Source Types
High Confidence (>60%)	16 numbers	11.0%	Official shareholder lists, tax portals (e.g. ComTax), direct registry matches.
Medium Confidence (30%–60%)	28 numbers	19.2%	Regional business directories matching subject name and city.
Low Confidence (<30%)	102 numbers	69.8%	Multi-page PDF documents (Scribd), B2B directories, broad web scrapings.

Top Data Sources & Domain Attribution

Domain	Matches	Primary Content Type	Quality Rating
scribd.com	53	IEPF dividend filings, court lists, gazette documents	High Recall / Variable Precision
tradeindia.com / indiamart.com	14	B2B supplier listings	Medium (Often returns vendor line)
in.marketscreener.com	10	Corporate dividend default notices	High Recall
pdfcoffee.com	6	PDF document search portal	Medium
comtax.up.nic.in	5	State Commercial Tax registrations	High Precision (80%+ score)

3. Strategic Recommendations for Discussion

🚀 Actionable Enhancements

- **1. Search Query Optimization:** Reduce search expansion from 15 queries per subject down to 4–5 precision queries (combining Name + Pincode + Address). This will cut search duration from **7.5 minutes down to ~1–2 minutes**.
- **2. Domain Noise Filtering:** Blacklist non-relevant search domains (e.g. Merriam-Webster, generic dictionary tools) to eliminate false positives.
- **3. Caller ID / Truecaller Post-Validation:** Run extracted candidate phone numbers through a Caller ID / Reverse Phone lookup API to verify that the registered account owner's name matches the target subject.
- **4. Batch Input Deduplication:** Add pre-run deduplication logic to skip duplicate input subjects (e.g. SANJAY KUMAR duplicated in CSV lines 13 & 14).